🎤 Presentation Page — Complete Component Guide

A deep-dive into the **Presentation** page of GenomeIQ — the in-app slide deck for demos and defenses. For each component this document covers:

- 🎯 **What it is** (plain-language description)
- 🧠 **What it tells you** (how to read the output)
- ⚙️ **How it works under the hood** (the algorithm)
- 🪜 **How to use it** (the user flow)
- 💡 **Practical tips** (when it's useful, when to be cautious)

📐 Page Layout Overview

```
┌──────────────────────────────────────────────────────────────────┐
│  Sidebar            │  Top header                                │
│  (navigation)       │ ───────────────────────────────────────────│
│                     │                                             │
│                     │  ┌─────────────────────────────────────────┐ │
│                     │  │ 1. Slide frame (gradient border)        │ │
│                     │  │                                         │ │
│                     │  │   Active slide content                  │ │
│                     │  │   (16:9 aspect)                         │ │
│                     │  │                                         │ │
│                     │  └─────────────────────────────────────────┘ │
│                     │                                             │
│                     │  ┌─────────────────────────────────────────┐ │
│                     │  │ 2. Slide controls                       │ │
│                     │  │   ← Prev   ● ● ● ● ● ● ● ● ● ● ● ●   Next →│ │
│                     │  │   "Slide N / 12 · Use ← → keys"         │ │
│                     │  └─────────────────────────────────────────┘ │
└──────────────────────────────────────────────────────────────────┘
```

The Presentation page is a **max-5xl-wide** centered column with two regions: a 16:9 slide frame at the top and a navigation bar below.

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1. 🎞️ Slide Frame (12 Slides Total)

🎯 What it is

The main content area showing one slide at a time. Wrapped in a brand-color gradient border (brand-600 → purple-600 → pink-600) for visual appeal. Each slide has a **fixed 16:9 aspect ratio** so it looks like a real PowerPoint frame.

🧠 What it tells you

The 12 slides cover the project end-to-end:

#	Title	Content
1	**Title slide**	Project name "GenomeIQ" + subtitle + "MD Naimur Rashid"
2	**Problem**	DNA → disease prediction is hard (long sequences, noisy data, BLAST limitations)
3	**Solution**	A multi-layer ML pipeline (4 boxes: ML models, explainability, OOD, RAG)
4	**Architecture**	ASCII flow diagram (frontend → FastAPI → models + RAG)
5	**Dataset**	Class counts (377 / 164 / 154 / 113), augmentation pipeline
6	**Models**	Three backbones table (TF-IDF, DNABERT-2, Ensemble)
7	**Results**	Performance metrics (CV F1 0.96, hand-curated 4/4)
8	**Explainability**	Mock saliency map demo
9	**OOD + Similarity**	Trust signals: OOD risk + nearest neighbours
10	**RAG Copilot**	Pipeline diagram of the chat retrieval flow
11	**Tech stack**	6-tile grid of all technologies used
12	**Closing**	"Thank you" + name

Each slide is **layout-coherent**: title in brand color at top, body content fills the middle with appropriate visualisations (text, tables, ASCII diagrams).

⚙️ How it works under the hood

Each slide is a `<div>` with `x-show="slide === N"` directives. Only the active slide renders.

```
<div x-show="slide === 0" class="flex-1 flex flex-col items-center justify-center
text-center p-12">
  <div class="text-7xl mb-4">🧬</div>
  <div class="text-5xl font-extrabold ...">GenomeIQ</div>
  ...
</div>
<div x-show="slide === 1" class="flex-1 p-12">
  ...
</div>
...
```

`slide` is an integer from 0 to 11 in Alpine state. Switching slides only toggles `x-show` — no DOM teardown — so transitions are instant.

🪜 How to use it

This is your demo/defense slide deck. Use it during:

- A presentation in front of your professors / committee

- A live walkthrough for evaluators
- A recording for the YouTube demo

💡 Practical tips

- The deck is **read-only** — content is hardcoded in `frontend/index.html`. To edit a slide, find the corresponding `<div x-show="slide === N">` block and modify it.
- The **gradient border** is purely decorative; it makes the deck look polished even on a default browser.
- All slide text is responsive but limits to **slide-frame's min-height of 480 px** so it works at most laptop resolutions. Project at 1920×1080 for the best look.

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2. 🎮 Slide Controls (Navigation)

🎯 What it is

The control row below the slide frame. Three navigation methods all do the same thing: change the active slide.

🧠 What it tells you

Element	Meaning
← **Previous button**	Goes back one slide. Disabled (30% opacity) on the first slide.
Dot indicators (12 dots)	Each dot represents one slide. The active slide's dot is wider and brand-colored. Click any dot to jump there.
Next → button	Goes forward one slide. Disabled on the last slide.
Bottom caption	*"Slide N / 12 · Use ← → keys"* — reminds users that arrow keys also work.

⚙️ How it works under the hood

Three input methods:

1. **Buttons** call `prevSlide()` and `nextSlide()`:

   ```
   nextSlide() { if (this.slide < this.slideCount - 1) this.slide += 1; }
   prevSlide() { if (this.slide > 0) this.slide -= 1; }
   ```

2. **Dot click** sets the slide directly:

   ```
   <button @click="slide = i - 1" :class="..."></button>
   ```

3. **Keyboard** is bound to the entire window in `init()`:

```
window.addEventListener("keydown", (e) => this.onSlideKey(e));
```

The handler:

```
onSlideKey(e) {
  if (this.page !== "slides") return;            // only on this page
  if (e.key === "ArrowRight" || e.key === " ") this.nextSlide();
  else if (e.key === "ArrowLeft") this.prevSlide();
}
```

So **only when you're on the Presentation page**, arrow keys and spacebar work as slide controls. They have no effect anywhere else.

🪜 How to use it

Mouse navigation

- Click ← **Previous** / **Next** → to step through.
- Click any **dot** to jump directly to a specific slide.

Keyboard navigation (recommended for demos)

- → **(right arrow)** or **Space** → next slide
- ← **(left arrow)** → previous slide

💡 Practical tips

- During a **live demo**, use keyboard navigation. It looks more professional than clicking.
- The dot navigation is useful when **presenting to a discerning audience** — you can jump straight to a specific slide if a question requires it (e.g., "show me the architecture").
- If you want **fullscreen mode**, press **F11** in your browser to enter true presentation mode.

🔄 The Presentation Page Flow (End-to-End)

```
User clicks "🎤 Presentation" in sidebar
        ↓
Alpine sets page = 'slides'
        ↓
Slide 0 (title) becomes visible
        ↓
User presses arrow key OR clicks Next OR clicks a dot
        ↓
slide variable updates (0–11)
        ↓
Old slide's x-show evaluates false → hidden
```

```
New slide's x-show evaluates true → visible
        ↓
Visual transition appears instant
```

No API calls, no asynchronous waits, no rendering delays. The entire deck is **client-side static**.

🧠 Cheat Sheet — Suggested Demo Script

When presenting GenomeIQ live, follow this 12-slide narration:

Slide	Title	Talking points (~30 sec each)
1	Title	"Hi, I'm Naimur. I built GenomeIQ — a DNA disease classifier."
2	Problem	"DNA is long, noisy. BLAST needs known references. We need a tool for any sequence."
3	Solution	"Hybrid ML: classical ensemble + transformer. Plus explainability and chat."
4	Architecture	"Frontend → FastAPI → models + RAG. Walk through the diagram."
5	Dataset	"808 raw sequences, 4 classes. Augmented to 22,848 with sliding windows."
6	Models	"Three backbones: TF-IDF (best F1), DNABERT-2 (transformer), Ensemble (most robust)."
7	Results	"0.96 macro F1 on cross-val. 4/4 on hand-curated test sequences."
8	Explainability	"Saliency map shows which bases drove the prediction."
9	OOD + Similarity	"Two trust signals: OOD detector + BLAST-lite nearest hits."
10	RAG	"Copilot retrieves curated facts via FAISS, then asks Gemini to ground its answer."
11	Tech stack	"Built with: scikit-learn, PyTorch, Transformers, FastAPI, FAISS, Tailwind, Plotly."
12	Thank you	"Questions?"

Total: ~6 minutes. Pad each slide as needed.

🛠️ Customizing the Deck

Want to change a slide?

1. Open `frontend/index.html`.
2. Find `<!-- PRESENTATION SLIDES -->`.
3. Each slide is a `<div x-show="slide === N">`. Edit the inner HTML.
4. **Save**. Reload the browser — changes appear instantly (uvicorn `--reload` watches the SPA too).

If you add or remove a slide:

1. Update the slide HTML.
2. Update `slideCount` in `frontend/static/app.js` (currently `12`).

📖 Related Documents

- **`README.md`** — high-level project overview, contains a similar narrative
- **`DOCS.md`** — comprehensive technical documentation; deeper than the slides
- **`Guide/ANALYZE_PAGE_GUIDE.pdf`** — detailed walkthrough of the predictor (Slide 7's content elaborated)
- **`Guide/COPILOT_PAGE_GUIDE.md`** — detailed walkthrough of the Copilot (Slide 10's content elaborated)

Part of the GenomeIQ per-page documentation series.